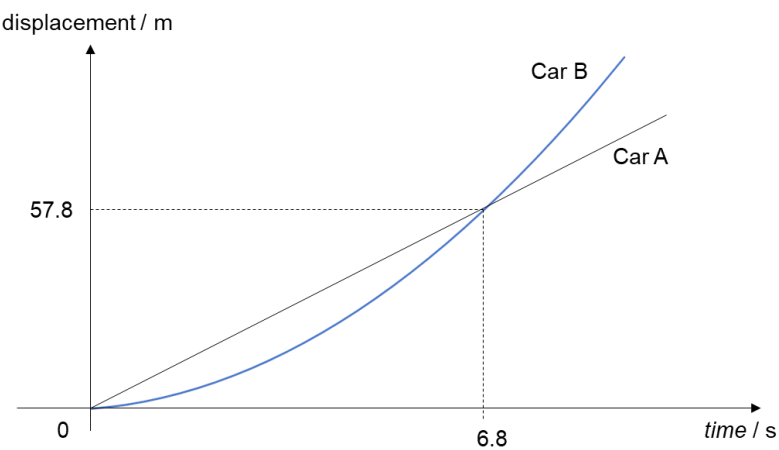


2)(a) (i)	$s_A = vt - (1)$ $s_B = ut + \frac{1}{2}at^2 = 0.5(2.5)(t^2)$ $s_A = s_B$ $8.5(t) = \frac{2.5}{2}(t^2)$ $t = 6.8 \text{ s}$
	$s_B = 0.5(2.5)(6.8)^2 = 57.8 \text{ m}$
(ii)	
(b) (i)	Vertical component $v_y^2 = u_y^2 + 2gs_y$ $v_y = \sqrt{2gs_y} = \sqrt{2(9.81)(6.5)} = 11.3 \text{ ms}^{-1}$
	$\tan \theta = \frac{11.3}{11.5}$ $\theta = 44.5^\circ$
(ii)	Final horizontal velocity is the smaller
	The other component of the velocity is correct θ will therefore be larger